

Area of continuous Time signal

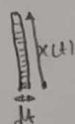
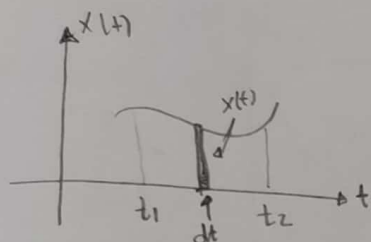
$$\text{Area of } x(t) = \int_{-\infty}^{\infty} x(t) \cdot dt$$

If $x(t)$ exist between t_1 and t_2 i.e

$$t > t_1 \Rightarrow x(t) = 0$$

$$t < t_2 \Rightarrow x(t) = 0$$

$$\text{Area of } x(t) \text{ over the Range } (t_1, t_2) = \int_{t_1}^{t_2} x(t) \cdot dt$$



$$\text{مساحة الشريط} = \text{الارتفاع} \times \text{العرض}$$

$$dA \times x(t) =$$

المجموع لجميع المسطحات = المساحة الكلية

$$\text{Area} = \int_{t_1}^{t_2} x(t) \cdot dt$$

E.X 8 compute the Area of $x(t)$ shown below:

$$\text{Area} = \int_{t_1}^{t_2} x(t) \cdot dt$$

$$m = \frac{\Delta y}{\Delta x}$$

الميل = التغير في y / التغير في x

$$\text{الميل } m = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow m = \frac{-1 - (1)}{2 - (-2)} \Rightarrow \frac{-2}{4} = -\frac{1}{2}$$

$$y = mx + c \Rightarrow 1 = -\frac{1}{2} \times -2 + c \Rightarrow 1 = +1 + c$$

$$c = 1 - 1 = 0$$

So

$$y = -\frac{1}{2}x \Rightarrow x(t) = -\frac{1}{2}t$$

$$\text{So the Area of } x(t) = \int_{-2}^2 x(t) \cdot dt \Rightarrow \int_{-2}^2 \frac{1}{2}t \cdot dt \Rightarrow \frac{1}{2} \cdot \frac{1}{2} [t^2]_{-2}^2$$

$$= \frac{1}{4} [4 - 4] \Rightarrow \text{the Area of } x(t) = \text{Zero}$$

$$A \triangleright = \frac{1}{2} \times 2 \times 1 \Rightarrow 1$$

$$A \triangleleft = \frac{1}{2} \times 2 \times -1 \Rightarrow -1$$

* إذا سلكنا المساحة الكلية، الناتج يساوي 0 لأن المساحة الموجبة والمساحة السالبة تلغى.

المساحة الكلية = 0

جميع المساحات يكون موجبة
 $A = -1 + (1)$
 $A = \text{Zero}$

$x(t)$: odd
 $\int_{t_1}^{t_2} x(t) dt = \text{Zero}$ if $t_1 = -t_2$ (2)

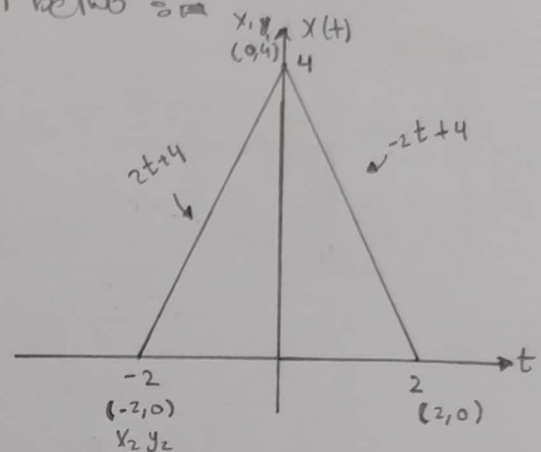
Example (2) = Compute the Area of $x(t)$ shown below =

$$m = \frac{\Delta y}{\Delta x} \Rightarrow \frac{0-4}{-2-0} = \frac{-4}{-2} \Rightarrow m=2$$

$$y = mx + C \Rightarrow 4 = 0 \times 2 + C \Rightarrow C = 4$$

So the $x(t) = 2t + 4$ $-2 \leq t \leq 0$

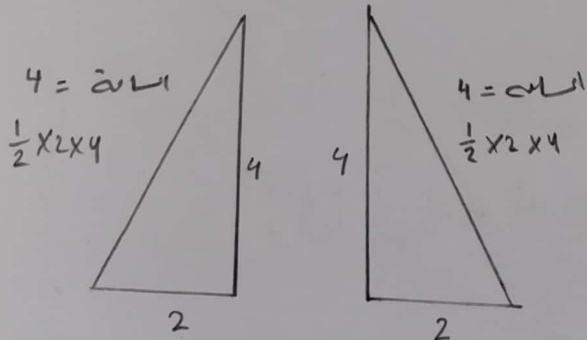
the same way $x(t) = -2t + 4$ $0 \leq t \leq 2$



$$\begin{aligned} \text{the Area} &= \int_{-2}^0 x(t) dt + \int_0^2 x(t) dt \\ &= \int_{-2}^0 (2t+4) dt + \int_0^2 (-2t+4) dt \\ &= \left[\frac{2}{2} t^2 + 4t \right]_{-2}^0 + \left[-\frac{2}{2} t^2 + 4t \right]_0^2 \end{aligned}$$

$$= 0 + 0 - [4 - 8] + [-4 + 8 - [0 + 0]] \Rightarrow 4 + 4 = 8$$

المساحة = $\frac{1}{2} \times \text{base} \times \text{height}$
 $= \frac{1}{2} \times 4 \times 4 \Rightarrow \text{Area} = 8$



او يمكن حساب مساحة مثلثات منفردة 2×4

$$\begin{aligned} \text{Area} &= \int_{t_1}^{t_2} x(t) dt \\ &= 2 \int_0^2 x(t) dt \end{aligned}$$

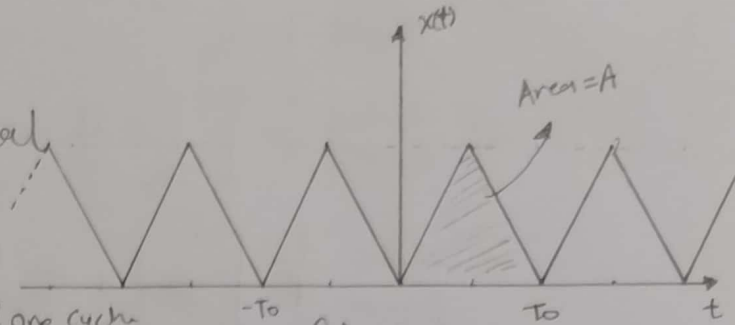
if $t_1 = -t_2$ for $x(t)$: even signal

"Average Value of Continuous time signal"

$$\text{Avg} = \frac{\text{total Area}}{\text{total time}}$$

← arbitrary

First Case if $x(t)$ is Periodic signal



In General $\text{Avg} = \frac{\text{total Area}}{\text{total time}} = \frac{\text{Area of one cycle}}{\text{time of one cycle}} = \frac{\text{Area of two cycle}}{\text{time of two cycle}}$

$$\text{Avg} = \frac{\int_{-\infty}^{\infty} x(t) \cdot dt}{\text{total time}} = \frac{\int_{T_0} x(t) \cdot dt}{T_0} = \frac{\int_{2T_0} x(t) \cdot dt}{2T_0} = \frac{A}{T_0}$$

So For Periodic signal

$$\text{Avg} = \frac{1}{T_0} \int_{T_0} x(t) \cdot dt$$

Second Case if $x(t)$ is non Periodic signal

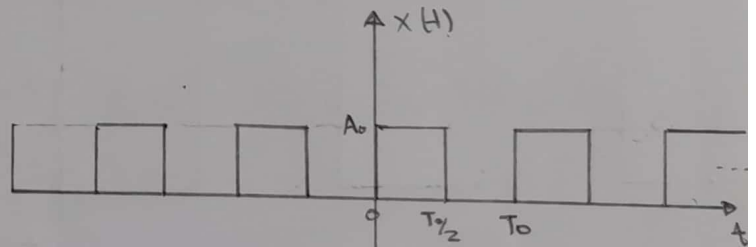
$$\text{Avg} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) \cdot dt$$

where T is the total time

EX Determine the Average of $x(t)$ show below.

Sol $\text{Avg} = \frac{1}{T_0} \int_{T_0} x(t) \cdot dt$

$$= \frac{1}{T_0} \int_0^{T_0/2} A_0 \cdot dt = \frac{1}{T_0} \cdot A_0 [t]_0^{T_0/2}$$



$$= \frac{A_0}{T_0} \cdot [T_0/2] \Rightarrow \text{Avg} = \frac{A_0}{T_0} \cdot \frac{T_0}{2} \Rightarrow \boxed{\text{Avg} = \frac{A_0}{2}}$$

OR $\text{Avg} = \frac{\text{Area of one cycle}}{\text{time period}} \Rightarrow \frac{A_0 \times T_0/2}{T_0} \Rightarrow \frac{A_0}{2}$

Ex 8 Compute the Average of $x(t)$ that shown below:

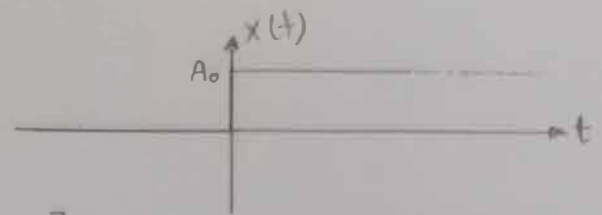
Soln the signal $x(t)$ is non-periodic

$$\text{Avg} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) \cdot dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \left[\int_{-T/2}^0 0 dt + \int_0^{T/2} A_0 dt \right]$$

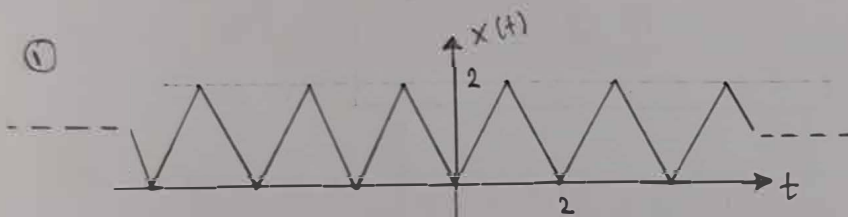
$$= \lim_{T \rightarrow \infty} \frac{1}{T} \left[\text{zero} + A_0 [t]_0^{T/2} \right] \Rightarrow \lim_{T \rightarrow \infty} \frac{1}{T} A_0 \cdot T/2$$

$$= \lim_{T \rightarrow \infty} \frac{A_0}{2} \Rightarrow \boxed{\text{Avg} = \frac{A_0}{2}}$$



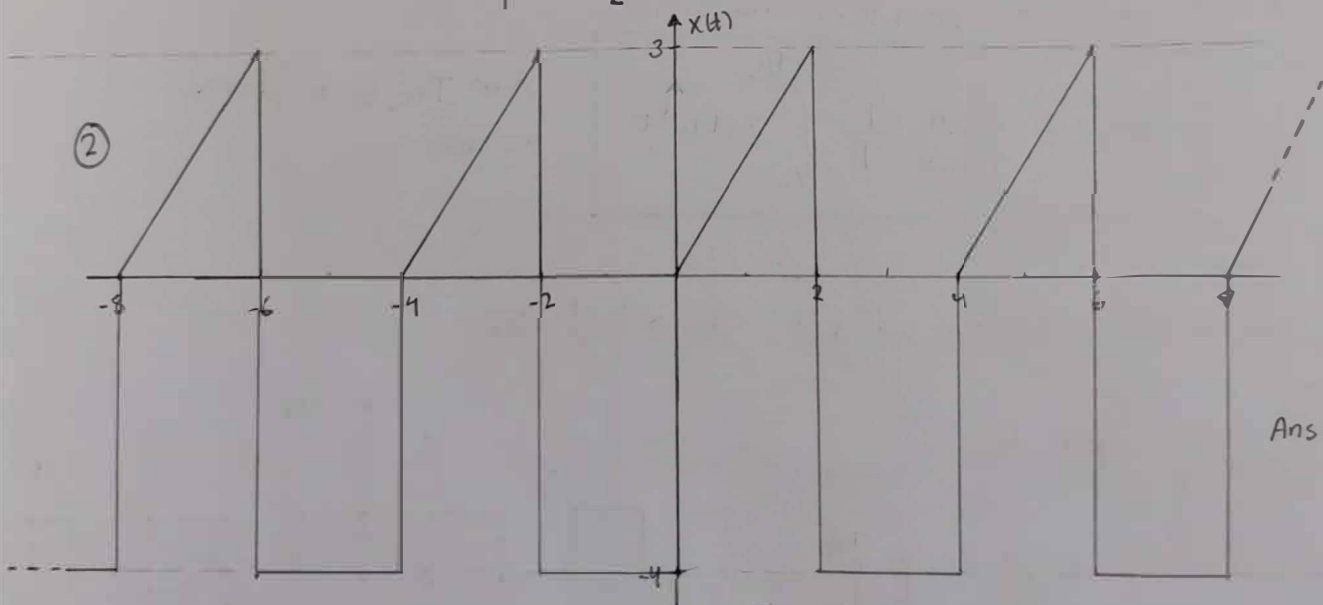
H.w Find the Average value of following continuous time signals:

①



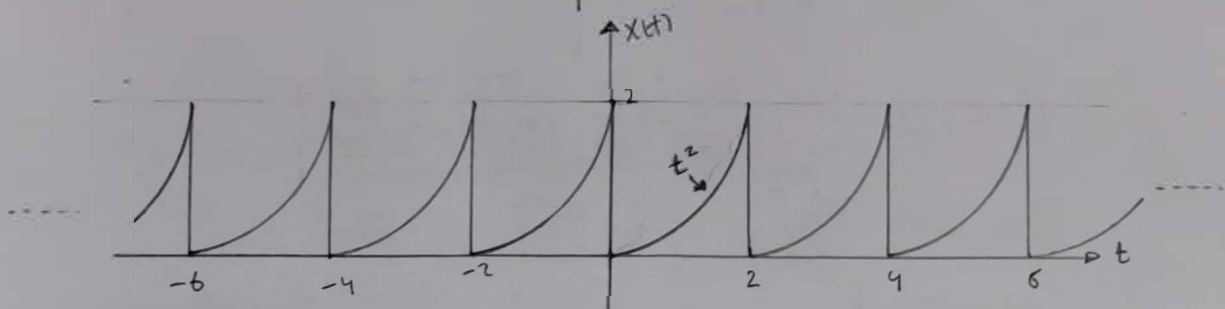
Ans: Avg = 1

②



Ans = $-\frac{5}{4}$

③



Ans = $4/3$

EX : Compute the Average Value for $x(t)$ = finite duration signal

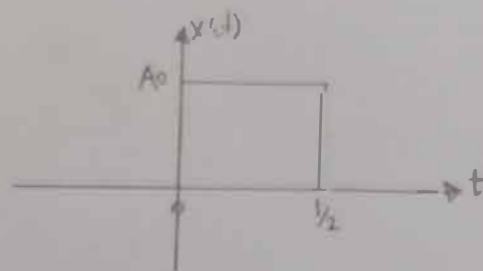
Sol :

$$\text{Avg.} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) \cdot dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{1/2} A_0 \cdot dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} A_0 [t]_0^{1/2} \Rightarrow \lim_{T \rightarrow \infty} \frac{1}{T} \cdot \frac{A_0}{2}$$

$$= \lim_{T \rightarrow \infty} \frac{A_0}{T \times 2} \Rightarrow \frac{A_0}{\infty} = \text{Zero}$$

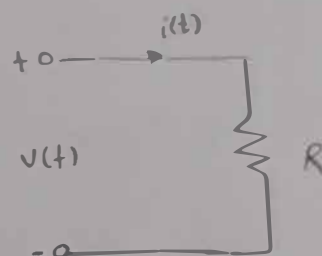


„Energy and Power of Continuous Time signals“

from Ohm's Law :

$$R = \frac{V(t)}{i(t)} \Rightarrow V(t) = i(t) \cdot R$$

$$i(t) = \frac{V(t)}{R}$$



the Power in R is $P(t) = i(t) \cdot V(t)$

$$= \frac{V(t)}{R} \cdot V(t) = \frac{V^2(t)}{R}$$

OR

$$= i(t) \cdot i(t) \cdot R = i^2(t) \cdot R$$

Let $R = 1 \Omega$ so $\Rightarrow P(t) = i^2(t) = V^2(t)$

IN General : Total Energy ; $\text{power} = \frac{\Delta E}{\Delta T} \Rightarrow E = \int P(t) \cdot dt$

$$E = \int_{-\infty}^{\infty} P(t) \cdot dt = \int_{-\infty}^{\infty} V^2(t) \cdot dt$$

$V(t)$: it is the voltage signal Can be written as $x(t)$

So

$$\text{Energy} = E = \int_{-\infty}^{\infty} |x(t)|^2 \cdot dt$$

the absolute Energy always positive .

$$\text{Avg. Power} = \frac{\text{total Power}}{\text{total time}}$$

⑥

$$\begin{aligned} & \xrightarrow{\text{For Periodic}} \frac{1}{T_0} \int_{T_0} P(t) \cdot dt = \frac{1}{T_0} \int_{T_0} |x(t)|^2 \cdot dt \\ & \xrightarrow{\text{For non-periodic}} \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} P(t) \cdot dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 \cdot dt \end{aligned}$$

Notes

$$\text{total Energy} = \int_{-\infty}^{\infty} |x(t)|^2 \cdot dt, \text{ For Periodic and non Periodic signals}$$

Avg. Power

$$P = \frac{1}{T_0} \int_{T_0} |x(t)|^2 \cdot dt, \text{ for Periodic signals}$$

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 \cdot dt, \text{ for non Periodic signals}$$

⇒ Avg. Power and Energy very important to find

- Energy Spectral Density.
- Power Spectral Density.
- Auto Correlation
- SNR . signal to Noise Ratio .

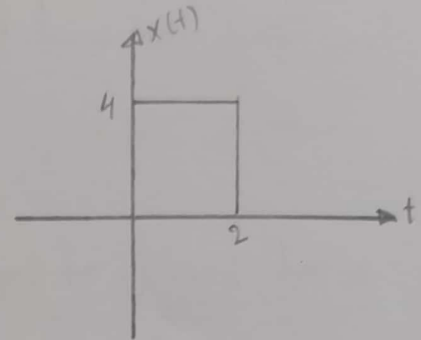
„Energy signals“

(7)

its Energy = Finite but Average Power = Zero

E.X Compute the Energy and Avg. Power for $x(t)$ shown below?

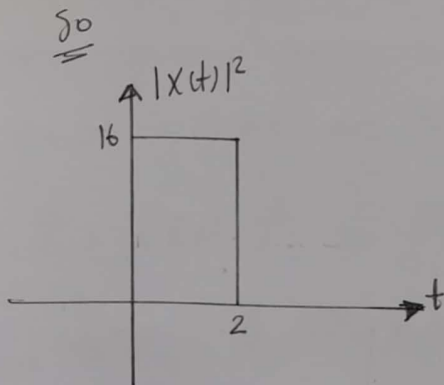
$$\begin{aligned} E &= \int_{-\infty}^{\infty} |x(t)|^2 \cdot dt \\ &= \int_0^2 (4)^2 \cdot dt \Rightarrow \int_0^2 16 \, dt \\ &= 16 \times 2 \\ E &= 32 \, \text{J} \end{aligned}$$



To Compute the Average Power $P = \lim_{T \rightarrow \infty} \frac{E}{T}$

So $P = 0$ because $\frac{32}{\infty} = \text{Zero}$

another way to Compute the Energy = Area of $|x(t)|^2$



$$\begin{aligned} \text{Area of } |x(t)|^2 &= \int_{-\infty}^{\infty} |x(t)|^2 \cdot dt \\ &= \int_0^2 16 \, dt \\ &= 16 [t]_0^2 \\ &= 16 \times 2 \\ E &= 32 \, \text{J} \end{aligned}$$

Properties of Energy signals:

(8)

1- Energy signals are absolutely integrable signals

$$\text{i.e. } \int_{-\infty}^{\infty} |x(t)| dt = \text{Finite} < \infty$$

and this the condition to compute Fourier transform for any signal.

2. total Energy of Energy signal = Area under $|x(t)|^2$ graph.

3- Usually the Average Power for Energy signals = Zero.

EX 8 Compute the Energy for $x(t)$ shown below?

$$x(t) = e^{-at} u(t) \text{ for } a > 0$$

Soln

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} (e^{-at} u(t))^2 dt$$

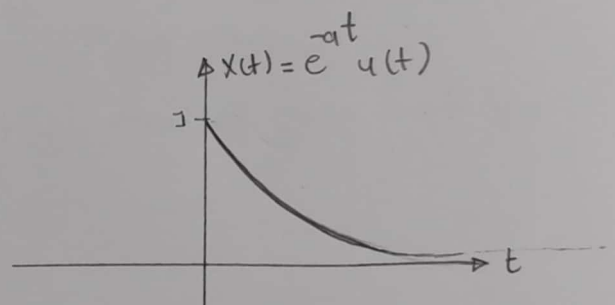
$$= \int_{-\infty}^{\infty} e^{-2at} \cdot u^2(t) dt$$

$$= \int_0^{\infty} e^{-2at} dt = \frac{1}{-2a} \left[e^{-2at} \right]_0^{\infty}$$

$$= \frac{1}{-2a} \left[\overset{0}{e^{-\infty}} - \overset{1}{e^0} \right]$$

$$E = 1/2a$$

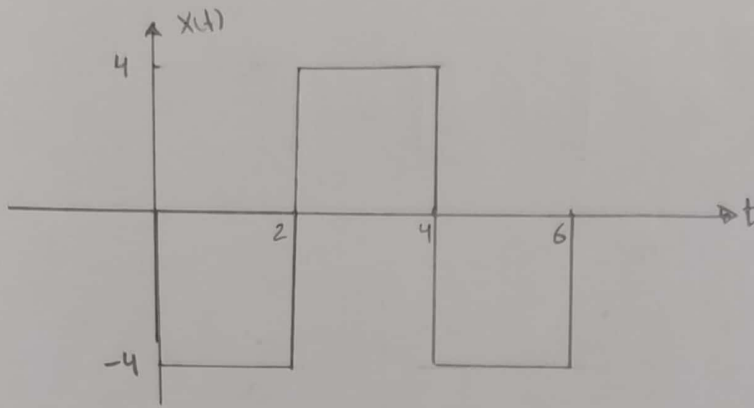
Average Power = Zero.



H.W Compute the Energy for $x(t)$ shown below?

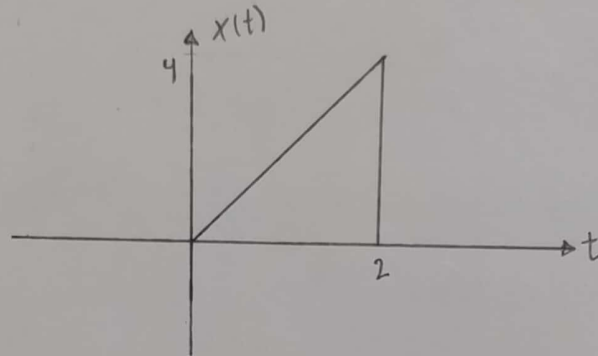
(9)

①



Ans: 96 J

②



Ans = $\frac{32}{3}$ J

"Power signals"

→ The Average Power for power signal always finite.

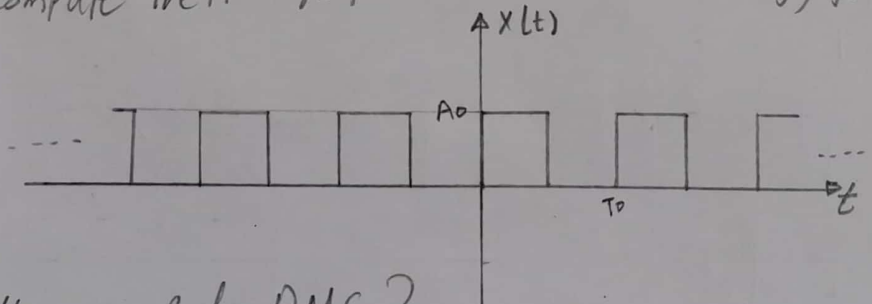
→ the Energy for Power signal = ∞ .

Properties of Power signals :

1- Periodic signals are Power signals but vice-versa is not true

$$2- P = (\text{RMS})^2 \Rightarrow \text{RMS} = \sqrt{P}$$

E.X : Compute the Average Power and total Energy for $x(t)$ show below



then compute RMS?

Sol

(10)

$$P = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |x(t)|^2 dt$$

$$= \frac{1}{T_0} \left[\int_{-T_0/2}^0 0 dt + \int_0^{T_0/2} A_0^2 dt \right] \Rightarrow \frac{1}{T_0} [A_0^2 t]_0^{T_0/2}$$

$$= \frac{1}{T_0} \times \frac{A_0^2 T_0}{2} \Rightarrow P = \frac{A_0^2}{2}$$

total Energy $\Rightarrow E = \infty$

$$RMS = \sqrt{P} = \sqrt{\frac{A_0^2}{2}} \Rightarrow \frac{A_0}{\sqrt{2}}$$

E.X Calculate average power of the following signal

$$x(t) = A_0 \sin \omega_0 t$$

Sol

$$P = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |x(t)|^2 dt \Rightarrow P = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} A_0^2 \sin^2 \omega_0 t dt$$

$$\boxed{\sin^2 \omega_0 t = \frac{1}{2} [1 - \cos 2\omega_0 t]}$$

$$P = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \frac{A_0^2}{2} [1 - \cos 2\omega_0 t] dt$$

$$P = \frac{A_0^2}{2T_0} \left[\int_{-T_0/2}^{T_0/2} 1 dt - \int_{-T_0/2}^{T_0/2} \cos 2\omega_0 t dt \right]$$

$$= \frac{A_0^2}{2T_0} \left[[t]_{-T_0/2}^{T_0/2} - \frac{1}{2\omega_0} [\sin 2\omega_0 t]_{-T_0/2}^{T_0/2} \right]$$

$$= \frac{A_0^2}{2T_0} \left[\left[T_0/2 - [-T_0/2] \right] - \frac{1}{2\omega_0} \left[\sin \left(2\omega_0 \times \frac{T_0}{2} \right) - \sin \left(2\omega_0 \times -\frac{T_0}{2} \right) \right] \right]$$

$$= \frac{A_0^2}{2T_0} \left[\left[\frac{T_0}{2} + \frac{T_0}{2} \right] - \frac{1}{2\omega_0} \left[\sin \left(\frac{2\pi}{T_0} \times \frac{T_0}{2} \right) - \sin \left(\frac{2\pi}{T_0} \times -\frac{T_0}{2} \right) \right] \right]$$

$$= \frac{A_0^2}{2T_0} \left[T_0 - \frac{1}{2\omega_0} \times 2 \sin(2\pi) \right] \quad \sin 2\pi = 0$$

$$P = \frac{A_0^2}{2T_0} \times T_0 \Rightarrow P = \frac{A_0^2}{2}$$

" Operations (Scaling, Shifting, Reversal) and Avg Power "

(11)

note :- time shifting

- Phase shifting

- time scaling

- Reversal time

- Amp. Reversal

these operations don't effect on
the value of average power

So if $x(t) = A_0 \sin \omega t$ then $P = A_0^2/2$

→ time shifting → $x(t \pm a) = A_0 \sin(\omega(t \pm a))$
 $= A_0 \sin(\omega t \pm a\omega)$ then $P = \frac{A_0^2}{2}$

→ phase shifting → $A_0 \sin(\omega t + \phi)$ then $P = \frac{A_0^2}{2}$

→ time scaling → $x(at) = A_0 \sin(\omega at)$ then $P = \frac{A_0^2}{2}$

→ Rev. time → $x(-t) = A_0 \sin(-\omega t)$ then $P = \frac{A_0^2}{2}$

→ Amp. Reversal → $-x(t) = -A_0 \sin(\omega t)$ then $P = \frac{A_0^2}{2}$

But Amp. Scaling

$x(t) \rightarrow A_0 \sin \omega t$ its Power $A_0^2/2$

$K x(t) \rightarrow K A_0 \sin \omega t$ $P = |K|^2 \cdot \text{Power of } (x(t))$

So $K x(t) = K A_0 \sin \omega t$ its Power equal to $\frac{K^2 A_0^2}{2}$

Amp. shifting

$$x(t) = A_0 \sin \omega t$$

$$x(t) + K \Rightarrow \text{the Power} = \frac{A_0^2}{2} + K^2$$

↑
DC signal

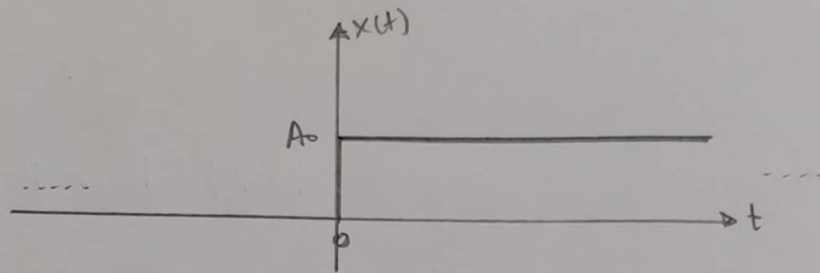
Ex If $x(t)$ is a power signal with $P=8$ what is the Power for $y(t) = -2x(3t-8)$

Soln Avg. Power of $y(t) = |k|^2 \cdot \text{Avg. Power of } x(t)$

$$= |-2|^2 \cdot 8$$

$$= 4 \times 8 \Rightarrow \text{Avg. Power of } y(t) = 32 \text{ watt}$$

Ex Compute the Average Power for non-periodic CTS shown below?



Soln $x(t)$ is non-periodic signal so

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

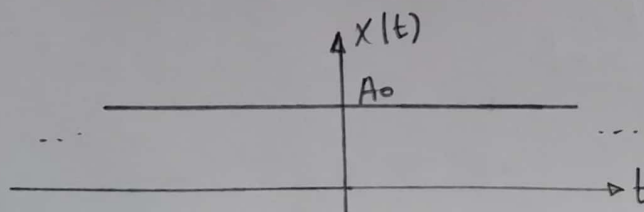
$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{T/2} A_0^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{A_0^2}{T} \left[t \right]_0^{T/2} \Rightarrow P = \lim_{T \rightarrow \infty} \frac{A_0^2}{T} \times \frac{T}{2}$$

$$P = \lim_{T \rightarrow \infty} \frac{A_0^2}{2} \Rightarrow \text{Average Power } P = \frac{A_0^2}{2}$$

$$RMS = \sqrt{P} = \frac{A_0}{\sqrt{2}}$$

H.W Compute the Average Power for DC signal shown below?



Ans: A_0^2

H.W Compute the Average Power for $x(t) = 5 \sin(10t) + 10 \cos(30t)$?

note:

$$A_0 \sin \omega_0 t \rightarrow P = \frac{A_0^2}{2}$$

Ans = 62.5 watt

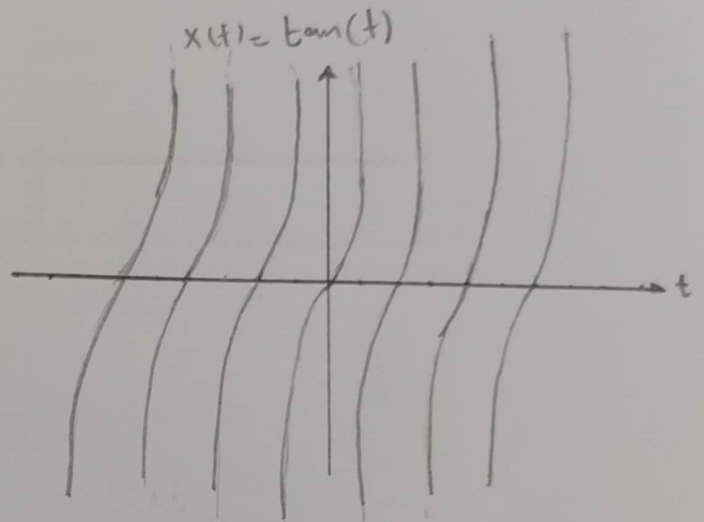
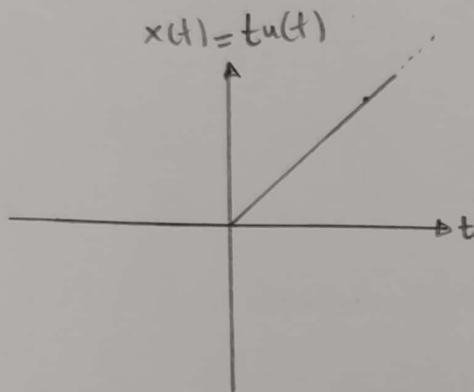
• Neither Energy Nor Power signal

(13)

"NENP"

if magnitude of signal is infinit at any instant of time then the signal will be neither energy nor power signal.

some examples on NENP signals:



$$\rightarrow x(t) = t^2$$

$$\rightarrow x(t) = 1/t$$

$$\rightarrow x(t) = e^t u(t)$$

$$\rightarrow x(t) = e^{-t} u(-t)$$

* Any Periodic signal is Power signal if all its values [magnitudes] don't equal to ∞ at any instant.